

NHANES Dataset Summary

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Introduction The National Health and Nutrition Examination Survey (NHANES) provides critical data on the health and nutritional status of the population (Centers for Disease Control and Prevention 2023).

Participant Characteristics The table below outlines the overall sample size, gender distribution, ethnic breakdown, and summary statistics for average systolic blood pressure across each survey wave.

```
# Calculate summary statistics by wave
summary_data <- nhanes %>%
  group_by(wave) %>%
  summarise(
    total_n = n(),
    male_n = sum(gender == "Male", na.rm = TRUE),
    male_pct = round((male_n / total_n) * 100, 1),
    white_n = sum(ethnicity_2 == "White", na.rm = TRUE),
    white_pct = round((white_n / total_n) * 100, 1),
    black_n = sum(ethnicity_2 == "Black", na.rm = TRUE),
    black_pct = round((black_n / total_n) * 100, 1),
    mexican_n = sum(ethnicity_2 == "Mexican", na.rm = TRUE),
    mexican_pct = round((mexican_n / total_n) * 100, 1),
    sbp_min = round(min(avg_sbp, na.rm = TRUE), 1),
    sbp_max = round(max(avg_sbp, na.rm = TRUE), 1),
    sbp_median = round(median(avg_sbp, na.rm = TRUE), 1),
    sbp_mean = round(mean(avg_sbp, na.rm = TRUE), 1),
    sbp_sd = round(sd(avg_sbp, na.rm = TRUE), 1)
  )

# Build the LaTeX-friendly table
summary_data %>%
  kable(
    col.names = c("Wave", "Total N", "N", "%", "White (N)", "White (%)", "Black (N)", "Black (%)", "Mexican (N)", "Mexican (%)", "Min", "Max", "Median", "Mean", "SD"),
    caption = "NHANES Sample Characteristics by Wave",
    booktabs = TRUE,
    align = "c"
  ) %>%
  kable_styling(latex_options = c("scale_down", "hold_position")) %>%
  add_header_above(c(" " = 2, "Males" = 2, "Ethnicity" = 6, "Systolic BP" = 5)) %>%
  row_spec(0, bold = TRUE)
```

As demonstrated in the table, the distribution of participants across major demographic categories and the average systolic blood pressure measurements remain relatively consistent across the recorded survey waves.

Centers for Disease Control and Prevention. 2023. "National Health and Nutrition Examination Survey."
<https://www.cdc.gov/nchs/nhanes/index.htm> (<https://www.cdc.gov/nchs/nhanes/index.htm>).